Project Design Phase-II: Technology Stack (Architecture & Stack)

Global Olympic Sports Performance Analysis (1992–2026)

Date: 31 January 2026	**Team ID:** NOC-PERF-2026
Project Name: Global Olympic Sports Performance Analysis	**Maximum Marks:** 4 Marks

Define Your Technology Stack

A robust business intelligence architecture selected to handle historical data processing across 60 countries, 50 sports categories, and multi-game Olympic datasets (1992–2026).

S.No	Component	Description	Technology	Justification / Purpose
1.	**Programming / Query Language**	Language used to query, filter, transform, aggregate, and analyze data.	**Python 3.11, SQL**	Python pandas/numpy enable complex statistical computations (per-athlete medal yield, regional efficiency ratios); SQL queries perform fast database aggregations.
2.	**Database / Data Warehouse**	Warehouse used to store and manage structured business data for analysis and reporting.	**Snowflake Data Warehouse / MySQL**	Provides scalable cloud storage for high-volume historical athlete records, medal tallies, and financial investment metrics spanning 1992 to 2026.
3.	**Data Ingestion**	Used to load data from source files or external systems into the analytical environment.	**Snowflake COPY INTO / Tableau Data Connections**	Automates batch ingestion of CSV/XLSX datasets (`athletes.csv`, `medal_results.csv`, `investment_data.csv`) into relational staging tables.
4.	**Data Transformation & Business Logic**	Tools/Logic used to clean, join, filter, aggregate data for analysis and calculate KPIs, metrics, and analytical measures.	**Tableau Prep Builder, Snowflake SQL Views**	Cleans demographic records, joins athlete listings with medal results, creates aggregated calculated fields (e.g., Avg Medals per Athlete, Regional Efficiency %).
5.	**Data Visualization / Analytics**	Used to create interactive dashboards, charts, reports, KPIs, and visual analysis.	**Tableau Desktop, Python (matplotlib/seaborn)**	Builds interactive horizontal bar charts, bubble charts (Investment vs. Medals), line graphs (1992-2026 medal trends), and donut charts for regional breakdown.
6.	**Front-End / User Interface**	Interface through which users interact with dashboards, filters, parameters, and visual insights.	**Tableau Public / Tableau Server, Streamlit**	Delivers self-service executive dashboards and narrative stories for NOC strategists, grant analysts, and media journalists.